

AIM: Calculate weight and balance for the training aircraft; explain the effects of CG position on handling and stability.

CALCULATION METHOD

- **Moment = Weight × Arm** (arm measured from datum in POH)
- Sum all weights → total weight
- Sum all moments → total moment
- **CG = Total Moment ÷ Total Weight**
- Plot CG at the calculated weight on the POH envelope chart or refer to forward and aft limits
- **Also check at zero fuel weight** – fuel burn can move CG out of limits

EFFECT OF CG ON HANDLING

- **Forward CG:** increased stability · higher stall speed · heavier stick forces · reduced cruise performance · there is a forward limit – elevator may be insufficient for flare
- **Aft CG:** reduced stability · lower stall speed in level but **dangerously** reduced stability at stall · risk of unrecoverable pitch-up – **the aft limit is the more critical limit to exceed**
- **AVGAS density:** 0.72 kg/L

POH LIMITS TO FIND

- ☐ Basic Empty Weight (BEW)
- ☐ Maximum Take-Off Weight (MTOW)
- ☐ Maximum Landing Weight (MLW)
- ☐ Maximum Baggage Weight
- ☐ Loading station arms (pilot, pax, baggage, fuel)
- ☐ CG forward limit and aft limit
- **Empty values come from the aircraft's own W&B report – cannot substitute another aircraft's report**

COMMON ERRORS

- **Forgetting zero fuel CG check** → Fuel burn shifts CG; check both MTOW loading and zero fuel loading
- **Wrong fuel density** → AVGAS = 0.72 kg/L
- **CG dismissed as “just within limits”** → limit is safety-critical; no margin should be accepted casually

STANDARDS FOR PROGRESSION

P

S

C standard required before flight test – no P tier for this lesson.

C

NC

C: All calculations correct without arithmetic error · CG correctly plotted at correct weight · All POH limits identified without assistance · Correctly describes forward vs aft CG handling effects · Identifies corrective action for out-of-limits scenarios

SAFETY Ground lesson only – no flight safety considerations

Application: From this lesson, W&B must be calculated for any non-standard loading and confirmed correct before the flight is authorised. Student calculates and briefs instructor before first flight after this lesson.